

8.4 Spectrum of noise and power spectral density

A primary goal of the communication system is to transmit more signal power against noise power to achieve greater signal-to-noise ratio (S/N). Noise, being random in nature, cannot be expressed as a time function, like deterministic waveform. It is represented by its average power. Hence, to evaluate the S/N, it is necessary to find a method to calculate the power of a random signal.

For a general deterministic signal $g(t)$, its power is given by

$$P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |g(t)|^2 dt$$

However, if $g(t)$ is random (such as noise), we can express its average power.

$$P_{avg} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \overline{|g(t)|^2} dt$$

8.4.1. The power spectral density of random signals

Let us define the Fourier Transform of a random signal $g(t)$ as $G(\omega)$. The power spectral density (PSD) $S_g(\omega)$ of $g(t)$ is defined as

$$S_g(\omega) = \overline{|G(\omega)|^2}$$

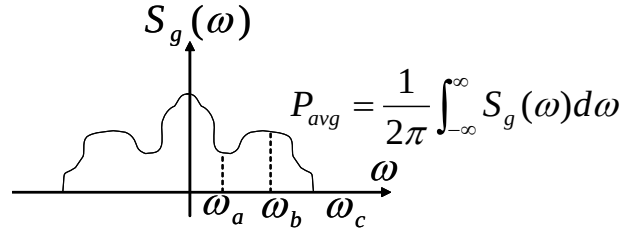
Properties of $S_g(\omega)$:

1) $S_g(\omega) > 0$ for all ω

2) By Parseval's relation, $P_{avg} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} \overline{|g^2(t)|} dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_g(\omega) d\omega$

Therefore, one can calculate the noise power through its PSD. In addition, the PSD actually tells you how much the average power P_{avg} is contributed by signals at different frequencies (or different *spectral components* of the signal).

For example, let's say we have a random signal $g(t)$ with the following PSD.



From the PSD, we can say that statistically, $g(t)$ contains a 'bigger' spectral component at frequency ω_b than at ω_a (or $|G(\omega_b)| > |G(\omega_a)|$ most of the time), and $g(t)$ does not contain any component at frequency ω_c .

8.4.2. PSD of random signals in LTI systems

We have learnt that in an LTI system characterized by impulse response $h(t)$ (and transfer function $H(\omega)$), the input spectrum $G(\omega)$ and output spectrum $Y(\omega)$ is related by

$$Y(\omega) = G(\omega)H(\omega)$$

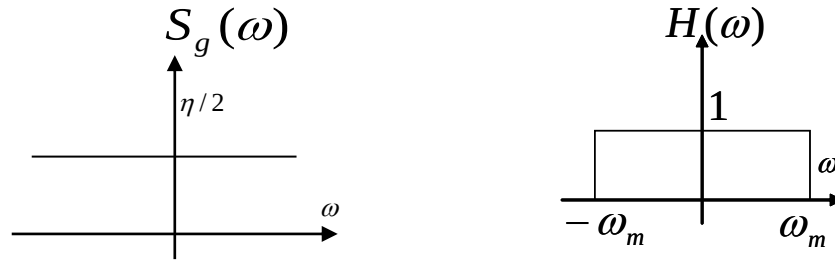
Now, if the input $g(t)$ is random with PSD $S_g(\omega)$, it can be shown that the PSD of the output signal $y(t)$ (also random since the input $g(t)$ is random) is given by

$$\begin{aligned} S_y(\omega) &= \overline{|Y(\omega)|^2} = \overline{|G(\omega)H(\omega)|^2} \\ &= \overline{|G(\omega)|^2} |H(\omega)|^2 \\ &= S_g(\omega) |H(\omega)|^2 \end{aligned}$$

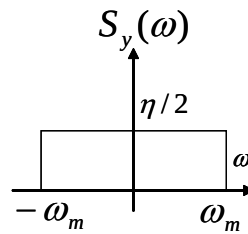
This is an important tool for communication system analysis since random noise generated along signal transmission will typically pass through a lot of LTI systems like filters and we would like to know its PSD at the end of transmission and consequently quantify its effect on the overall communication system performance

Example 8.5

Let $g(t)$ be a random signal with PSD $S_g(\omega)$ and it pass through a low-pass filter with transfer function $|H(\omega)|^2$, which are shown in the following figures.



In this case, the PSD of the output $y(t)$ will be $S_y(\omega) = S_g(\omega)|H(\omega)|^2$ and is given in the following figure.



The average power of $y(t)$ will be given by $P_{y,avg} = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_y(\omega) d\omega = \frac{\omega_m \eta}{2\pi}$

This type of PSD where the noise power is uniformly distributed across frequency $S_g(\omega) = \eta / 2$ is the most common form of noise in practice and they are known as *white noise*.

8.5 Error control coding

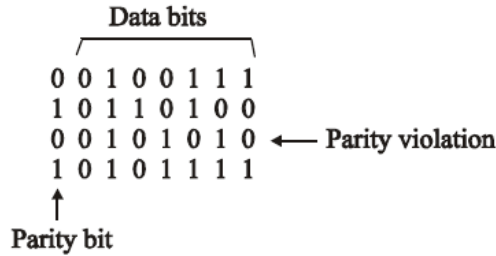
In *error control coding*, the *extra* bits are introduced into the transmitted signal to provide carefully structured redundancy, in order to detect the presence of errors in the received pattern. This idea is not new – we are using redundant digits and information to ensure reliability in our daily lives all the time. For example, our HKID card has an error detection feature.



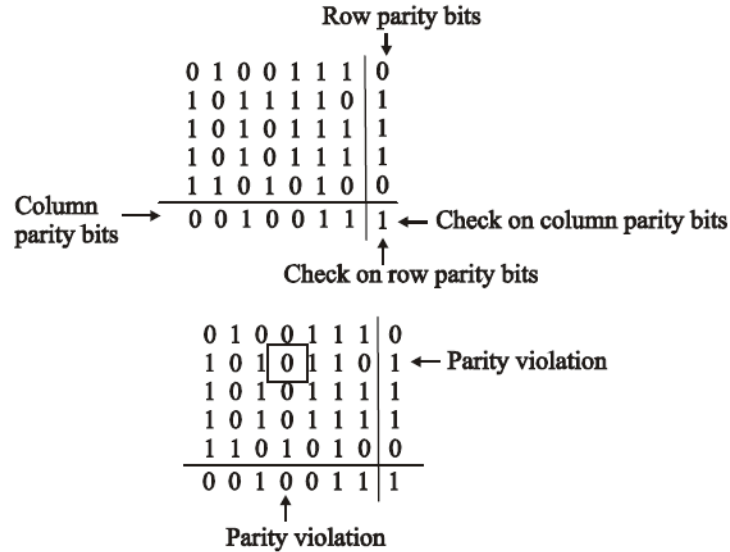
ID number	R=18	8	1	6	8	6	5	1
Weight	8	7	6	5	4	3	2	1
ID number x weight	144	56	6	30	32	18	10	1
Total sum								297 (=27x11) -> divisible by 11

Figure 8.13. The HKID number is arranged such that the total sum must be divisible by 11, thus creating an error detection mechanism for single digit error.

8.5.1. Parity checks for error detection



8.5.2. Row and column parity check



With such error detection however, we are not able to identify which bit is transmitted in error, and the receiver will ask the transmitter to re-transmit the data again. A more power strategy is to use error detection and correction codes.

8.5.3. Repetition code

Repetition code simply repeats each bit 3 times and transmit 3 copies of the same bit. At the receiver, it simply looks at the detected 3-bit sequence and decide the original bit based on whether there are more 1's or 0's in that 3-bit sequence. In this case, the *code rate* R is defined as the ratio of the original bit and the length of the coded bit sequence, or *codeword*. For the 3-bit repetition code, the code rate $R=1/3$. The advantage here is that with such coding scheme, the ultimate detection error probability $P(\hat{s} \neq s) = 3p^2 + p^3$ where p is the BER of a single detection error between the transmitted bit sequence and the received bit sequence i.e. $p = P(\mathbf{r} \neq \mathbf{t})$. For typical values of p , $3p^2 + p^3$ will be much smaller than p , thus the ultimate detection error probability will be significantly reduced. The drawback of such scheme is that the total bit rate will be reduced by the code rate R .

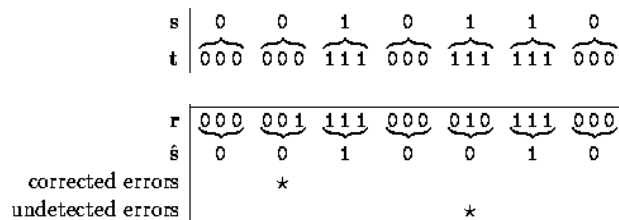


Figure 8.14. Repetition code with code rate 1/3. s: original bit sequence; t: transmitted codeword sequence; r: received bits with some detection errors; ŝ: recovered original bit sequence.

8.5.4. Hamming code

The goal of Hamming codes is to create a set of parity bits such that a single-bit error in a data bit or a parity bit can be detected and can be corrected. The following diagram shows a Hamming 7,4 code with 4 information bits d_1, d_2, d_3, d_4 and parity bits p_1, p_2, p_3 . In this case, the parity bits are given by

$$p_1 = d_1 + d_2 + d_4 \pmod{2}$$

$$p_2 = d_1 + d_3 + d_4 \pmod{2}$$

$$p_3 = d_2 + d_3 + d_4 \pmod{2}$$

Basically, the parity bits made sure that the group of 4 bits in each circle sums to 0 mod 2 (or even parity). The transmitted codeword is the 7-bit sequence $d_1, d_2, d_3, d_4, p_1, p_2, p_3$. In this case, we can show that if there is a detection error occurred to one of the bits, we can correctly detect and recover it.

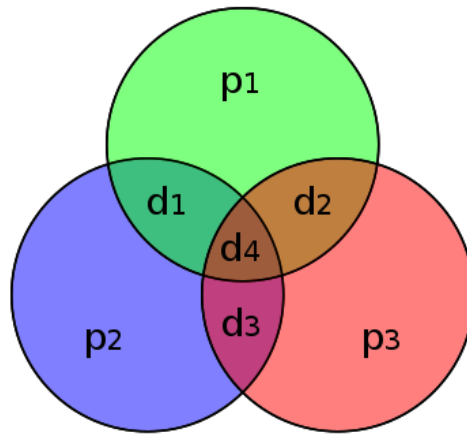


Figure 8.15. Hamming 7,4 code with code rate 4/7. The parity bits p_1, p_2, p_3 are designed such that the sum of bits in each circle is 0 mod 2 (or even parity). Any single detection error in one of this 7 bits can be identified and corrected.

Chapter 9 Optical Fiber Communications

Objectives

- To discuss the importance of optical fiber communication
- To introduce optical fiber communication system
- To describe the principle of laser
- To explain the working principle of photo-detector
- To illustrate light propagation in optical fibers
- To explain total internal reflection
- To introduce the concept of numerical aperture
- To study the transmission properties of optical fibers

The objective of any communications system is the transfer of information from one point to another. This information transfer is accomplished most often by superimposing (modulating) the information onto an electromagnetic wave (carrier). The modulated carrier is then transmitted (propagated) to the destination, where the electromagnetic wave is received and the information recovered (demodulated). Such systems are often designated by the location of the carrier frequency in the electromagnetic spectrum. In radio systems, the electromagnetic carrier wave is selected with a frequency from the radio frequency (RF) portion of the spectrum. In an optical fiber communication system, the carrier is selected from the optical region, which includes the infrared and visible frequencies.

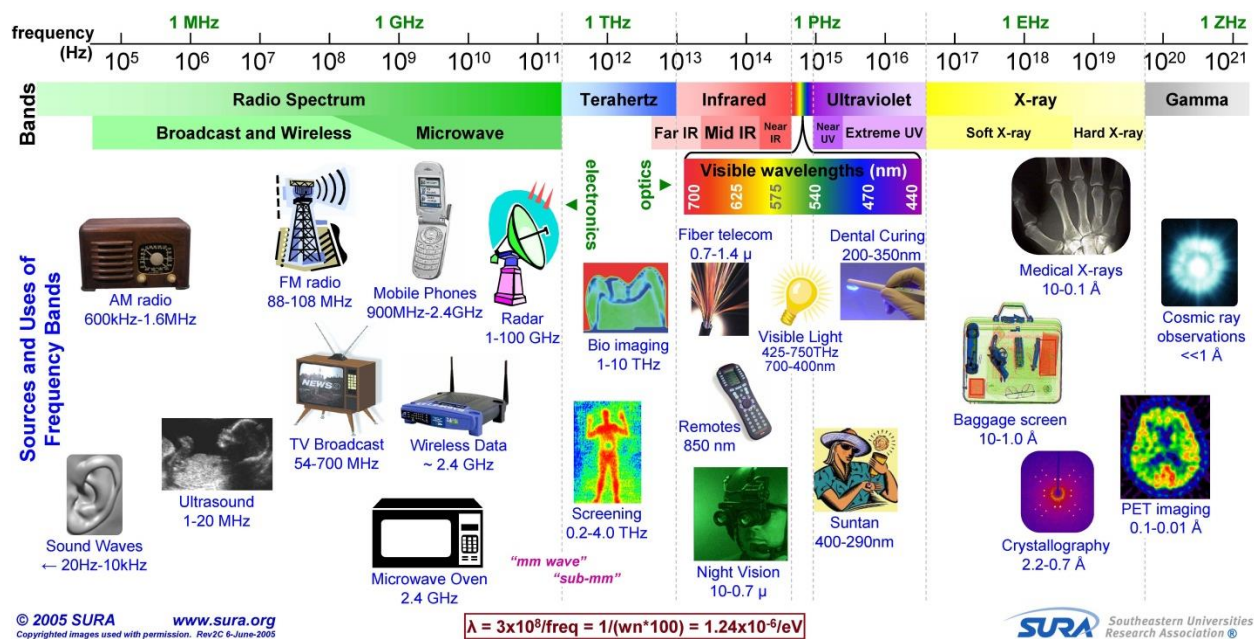


Figure 9.1. The electromagnetic spectrum.

What are the advantages of optical fiber communication system?

1. Wide bandwidth

In any communication system, the amount of information transmitted is directly related to the bandwidth of the modulated carrier. Thus increasing the carrier frequency increases the available transmission bandwidth, and therefore the information capacity of the overall system. The frequencies in the optical range will have a usable bandwidth approximately 10^5 times that of a carrier in RF range.

2. Low loss

Optical fibers have lower transmission losses than copper cables. In a copper cable, the attenuation increases with modulation frequency: the higher the frequency, the greater the loss.

Bandwidth is an indication of the rate at which information can be sent.

Loss indicates how far the information can be sent.

The combination of high bandwidth and low loss of optical fiber communication system means more data can be sent over longer distances, thereby decreasing the number of wires and the repeaters required. This reduction in equipment and components decreases the system cost and complexity.

3. Small size and weight
A fiber optical cable with the same information carrying capacity as copper cable weighs less than the copper cable because the copper requires more lines than the fiber.
4. Immunity to electromagnetic interference (EMI)
Unlike copper cables, optical fibers do not radiate or pick up electromagnetic radiation. This comes from the dielectric nature of optical fibers.
5. Electrical isolation
An optical fiber is made of glass, which is an electrical insulator, it creates no spark or fire hazard, and does not attract lightning.
6. Signal security
Optical signal is well confined within the waveguide, this provides a high degree of data security.
7. Abundant raw material
Silica is the principle material of which optical fibers are made. This raw material is abundant and inexpensive, since it is found in ordinary sand.

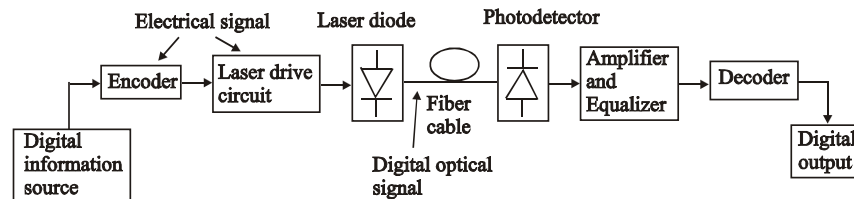


Figure 9.2. An optical fiber communication system uses a digital communication scheme.



Figure 9.3. Optical fiber communication systems using intensity-modulation. Bits “1” and “0” are represented, respectively, by the presence and absence of an optical pulse.

We will now talk about transmission properties of optical fibers. The most important transmission properties of optical fiber are **attenuation** and **dispersion**.

9.1 Attenuation

The various loss mechanisms in an optical fibre are:

- Material absorption
- Rayleigh scattering
- Bending loss (radiation losses)

9.1.1. Absorption

Absorption is a result of three mechanisms:

1. Absorption by atomic defects
2. Absorption by impurity atoms (mainly water)
3. Absorption by basic constituent atoms (Si, O, Ge)

Effect of OH ion absorption:

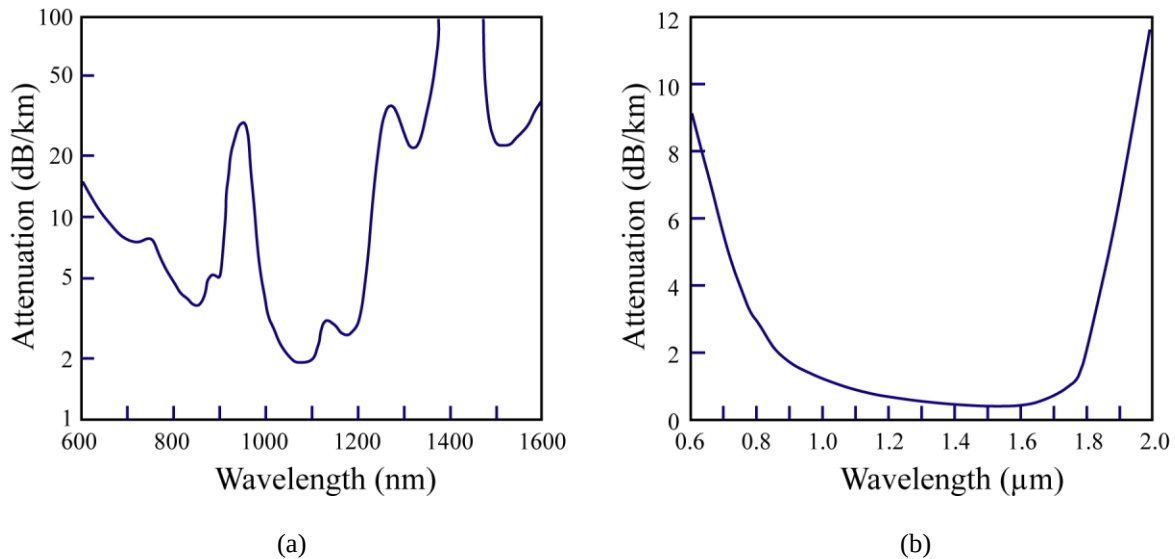


Figure 9.4. Attenuation vs. wavelength for a single-mode optical fibre. (a) fibre with a high water impurity content; (b) fibre with an OH content of less than 0.8 ppb.

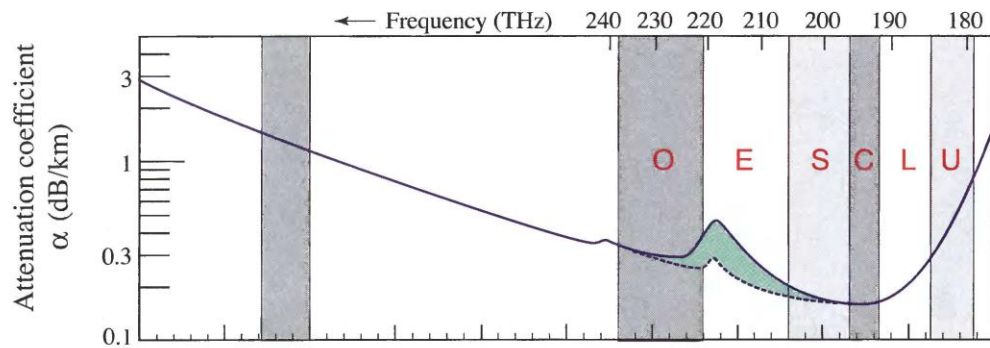


Figure 9.5. Attenuation vs. wavelength for a modern fibre.

Dopants used to produce the index step lead to different losses. For operation at longer wavelength, a GeO_2 doped fibre gives the lowest loss. Environmental effects (such as exposure to H_2 at high pressure, and to nuclear radiation) can induce greater absorption.

Hydrogen can affect the performance of optical fibres in many ways. Figure 9.4 (a) shows the induced loss of fibre at various wavelengths due to the absorption of molecular hydrogen in the fibre. The hydrogen does not react with the fibre and once the hydrogen is allowed to diffuse out of the fibre, the induced absorption peaks disappear.

In the second-generation case, however, the hydrogen reacts with the existing defect sites of the fibre and induced permanent absorption peak at 1383 nm as shown in Figure 9.4 (a). The 1530 nm absorption is transient and the 1240 nm absorption is the first overtone of the unreacted hydrogen.

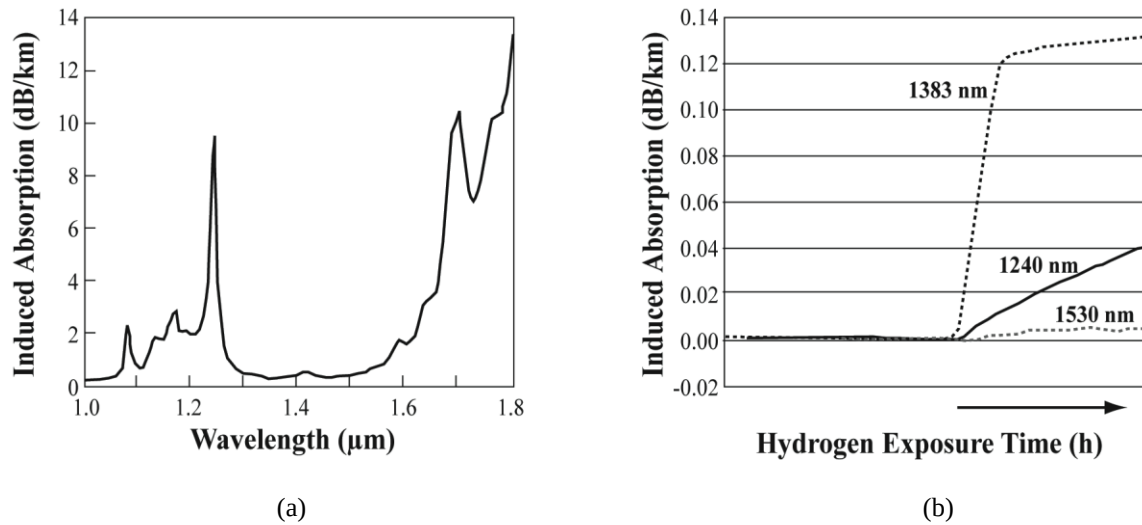


Figure 9.6. (a) Measured spectral absorption for fibre exposed to 1-atm hydrogen for 23 days at 15°C [IEEE Journal of Lightwave Technology, 1985] and (b) Second-generation hydrogen aging due to reaction of hydrogen with existing defect sites in the glass structure [Corning WP9007, 2004].

9.1.2. Rayleigh scattering

Rayleigh scattering arises from local microscopic fluctuations in glass density (composition), and from structural inhomogeneities. These may lead to random variations in the refractive index on a scale smaller than λ . Light scattering under these circumstances is called Rayleigh scattering. The intrinsic loss of silica fibres from Rayleigh scattering is

$$\alpha_R = \frac{c}{\lambda^4}$$

where c is the range 0.7 - 0.9 (dB/km) μm^4 depending on the constituents of the fibre core. At 1.55 μm , $\alpha_R = 0.12$ -0.16 dB/km.

Other inhomogeneities, having a scale larger than λ , such as gas bubbles and crystallized regions, cause Mie scattering which is negligible in today's fibres because of the high level of manufacturing expertise.

The decrease of α_R in proportion to λ^{-4} explains why halide fibres potentially have very low losses at mid-infrared wavelengths because their absorption losses in the infrared region also start to grow at wavelengths larger than 1.6 μm .

Attenuation limits how far a signal can travel through a fiber before it becomes too weak to be detected.

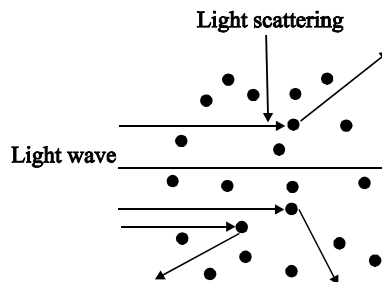


Figure 9.7. Illustration of scattering.

9.1.3. Attenuation coefficient

The variation of optical power with distance in a fibre:

$$\frac{dP}{dz} = -\alpha_p P$$

α_p [km⁻¹] is the attenuation coefficient which accounts for all power independent losses.

$$P_{out} = P_{in} e^{-\alpha_p L}$$

$$\alpha \text{ [dB/km]} = \frac{10}{L} \log_{10} \left(\frac{P_{in}}{P_{out}} \right) = 4.343 \alpha_p \text{ [km}^{-1}\text{]}$$

9.2 Dispersion

Dispersion is the spreading of a light pulse as it travels along an optical fiber. It occurs because the speed of light through a fiber depends on its wavelength and the propagation mode. Dispersion limits digital transmission speed by causing pulses to overlap, so they cannot be distinguished. The bit rate must be low enough to ensure that pulses do not overlap. A lower bit rate means that the pulses are farther apart and, therefore, that greater dispersion can be tolerated. A fiber with dispersion can be characterized as a linear and time-invariant system with transfer function

$$H(\omega) = e^{-j\omega^2 \beta_2 L/2}$$

where β_2 is called the group velocity dispersion (GVD) parameter of the fiber.

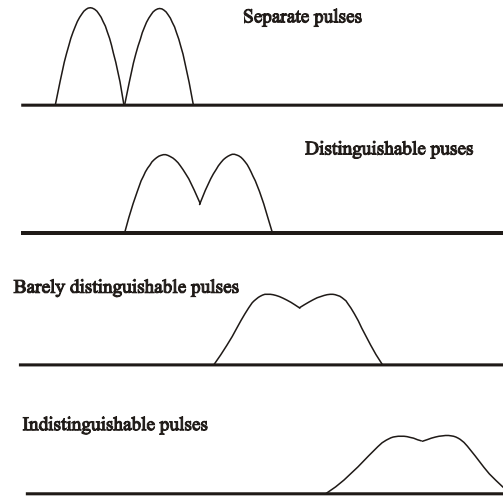


Figure 9.8. Illustration of pulse spreading caused by dispersion.

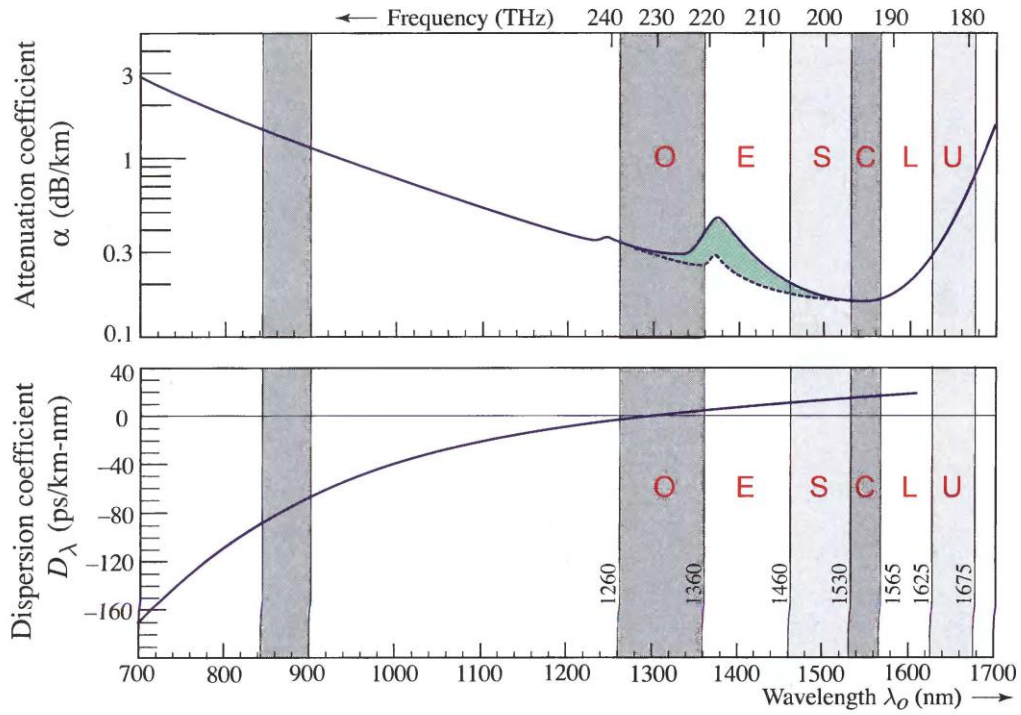


Figure 24.1-2 Wavelength dependence of the minimum attenuation α and the material dispersion coefficient D_λ of silica-glass fibers. The dashed line represents the attenuation of silica-glass fibers with suppressed OH absorption. Three spectral bands (shaded) are noted: the band centered at 870 nm, which was used in earlier systems has $\alpha = 1.5$ dB/km and $D_\lambda = -80$ ps/km-nm; the O (original) band centered at 1310 nm, for which $\alpha = 0.3$ dB/km and dispersion is minimal; and the C (conventional) band centered at 1550 nm, for which attenuation is minimal ($\alpha = 0.16$ dB/km) and $D_\lambda = +80$ ps/km-nm. Three additional bands are used in wavelength division multiplexing systems (WDM): E = extended, S = Short, L = Long; and U = Ultra-long.